

Passive Acoustic Monitoring Applications for Fish

Xavier Mouy, Rebecca Van Hoeck

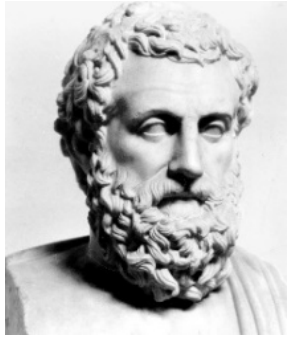


WOODS HOLE
OCEANOGRAPHIC
INSTITUTION



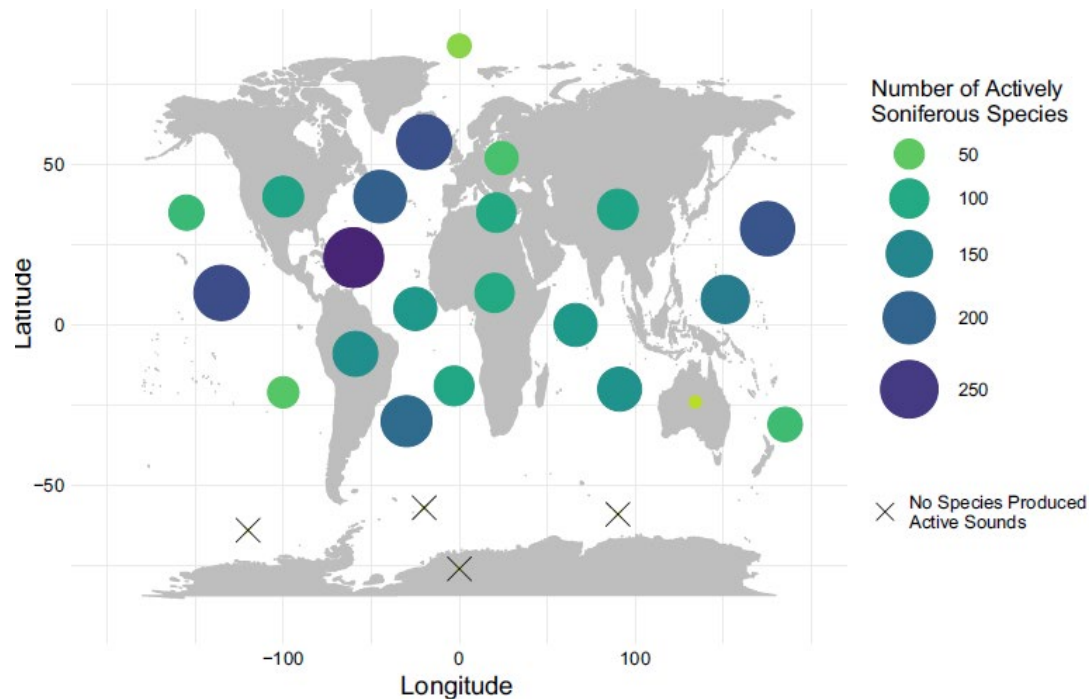
PAM SI Virtual Workshop
September 24 & 25th, 2026

Fish produce sounds



Aristotle 4th century BC:

“Fishes can produce no voice, for they have no lungs, nor windpipe and pharynx; but they emit certain inarticulate sounds and squeaks.”



Looby et al. 2022:

“989 fish species from 133 families and 33 orders have been shown to produce active (i.e., intentional) sounds”

Can PAM be useful for fisheries and conservation?

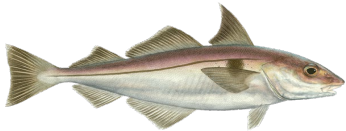
What we want...



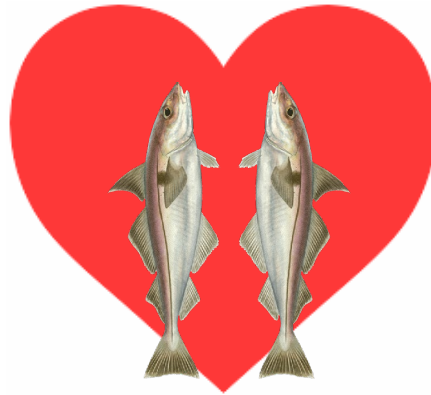
Presence



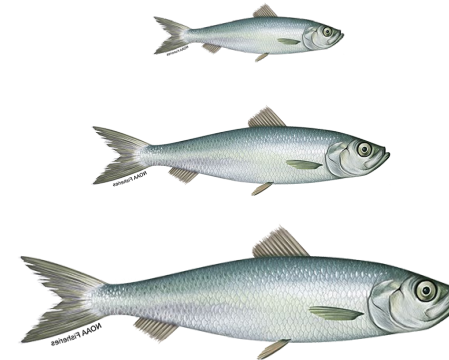
Species



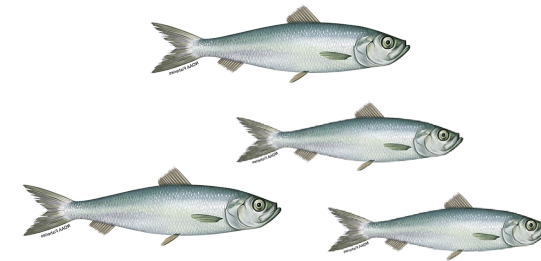
Behavior



Size



Count



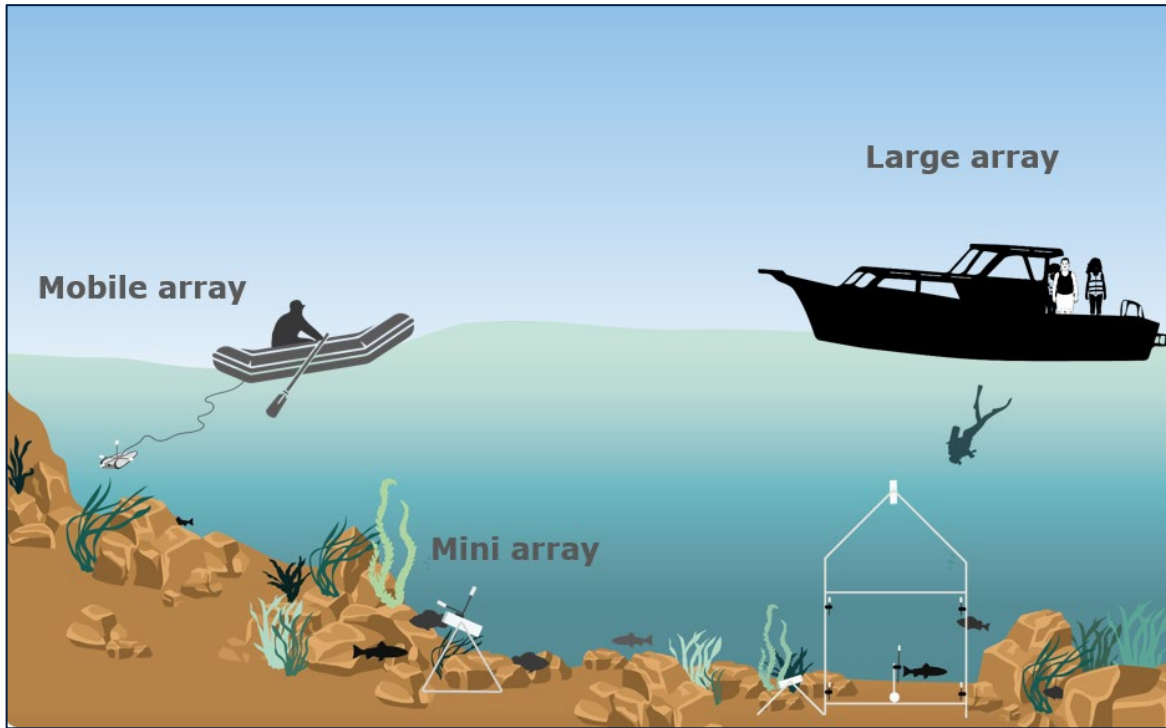
Complexity



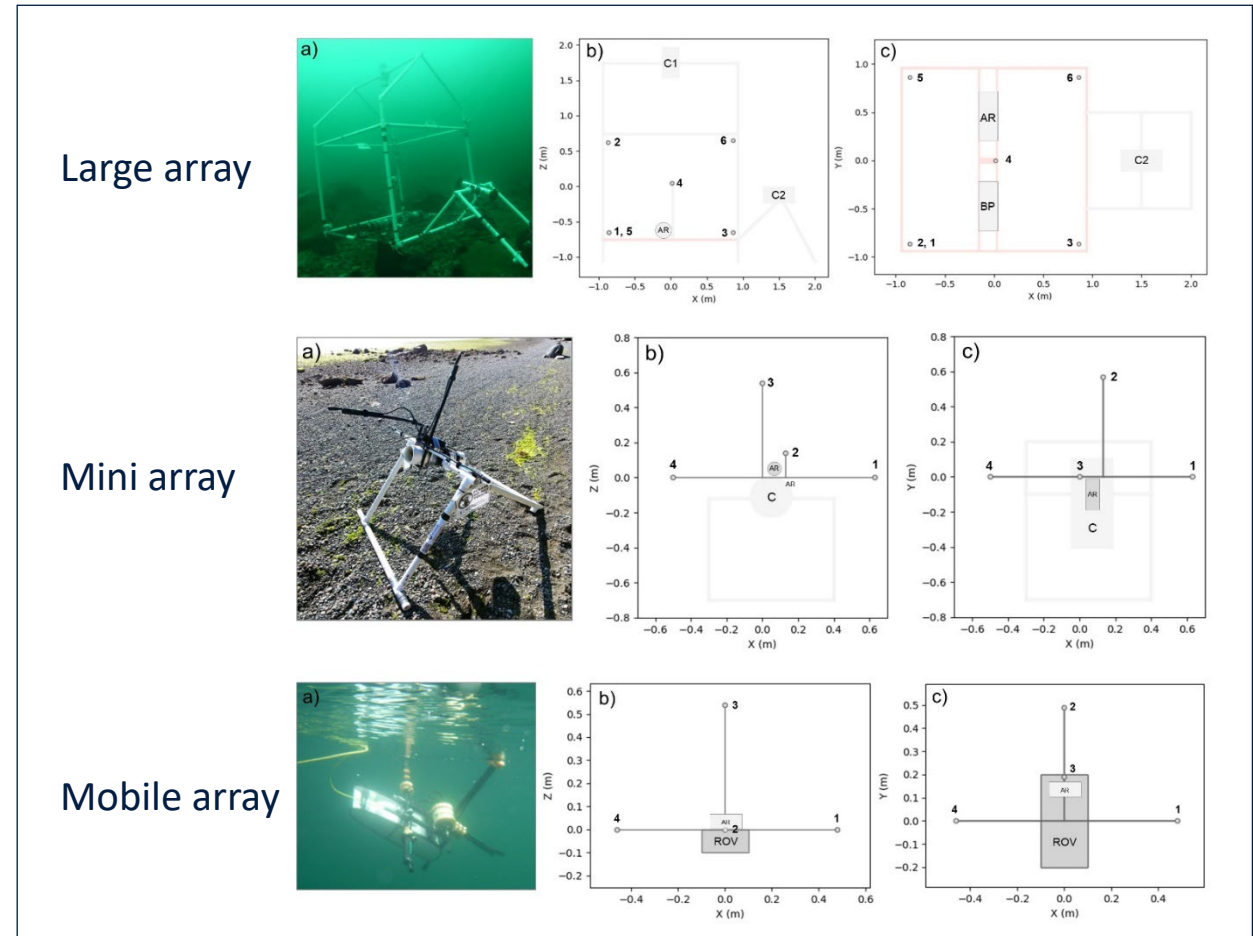
Species identification in the field

Audio-video arrays to study fish sounds in the wild

A set deployable solutions



Mouy et al., 2023



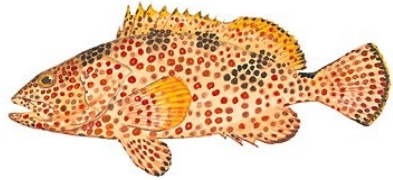
| Species identification in the field



Species identification in the field



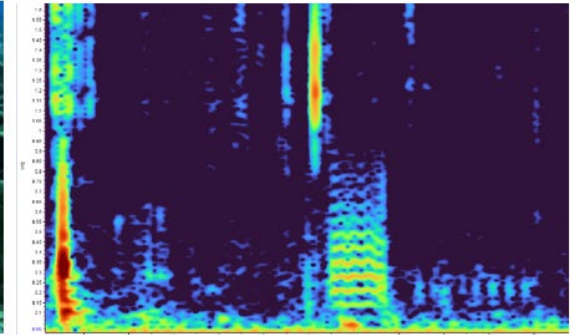
Diver-based array



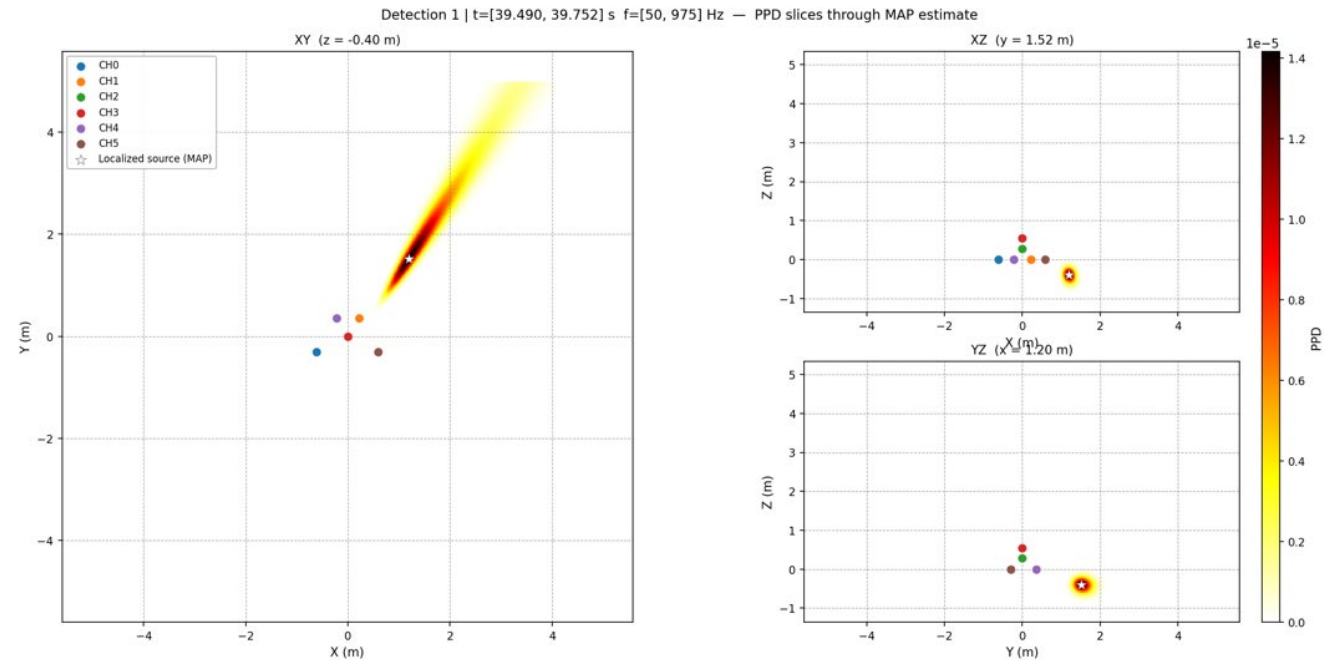
Camera (red hind seen in front right)



Grunt recorded

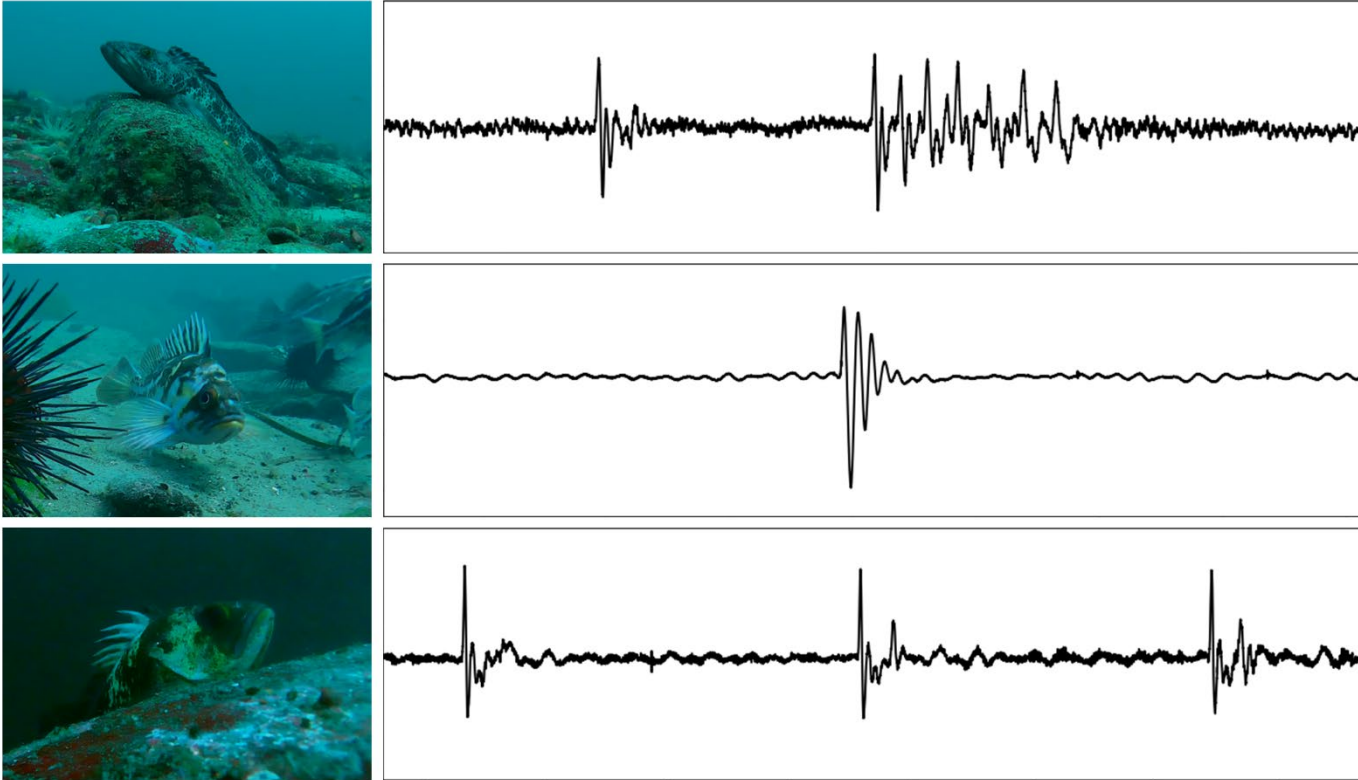


Localized grunt

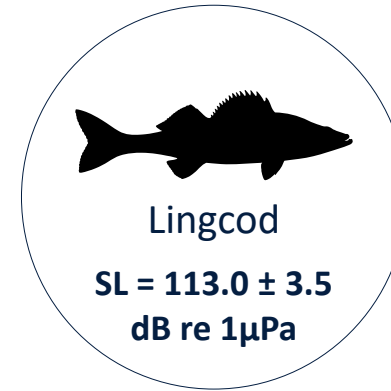


Species identification in the field

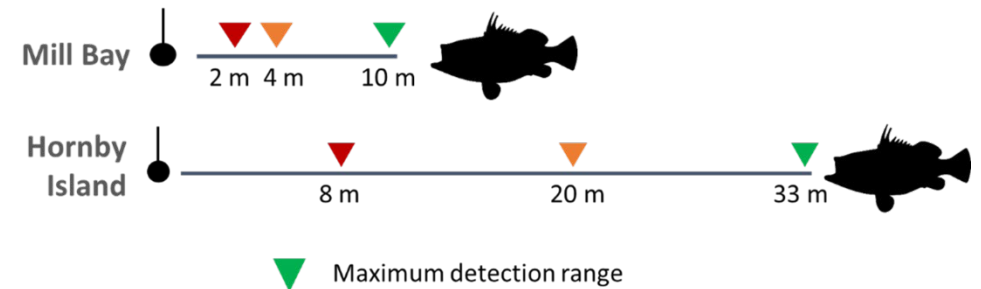
Sound description



Source levels



Detection range estimates



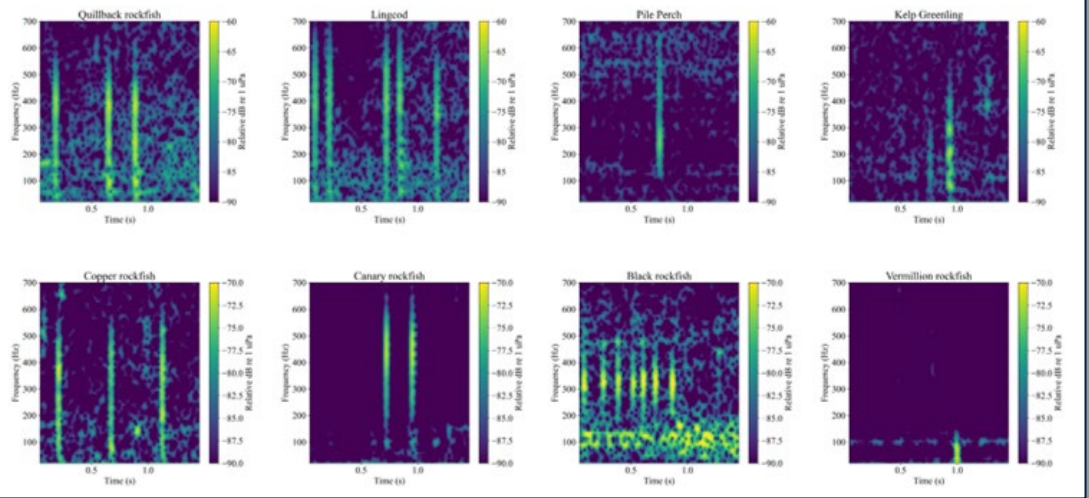
Species identification in the field



Identified a total of 1,107 fish calls for eight different species

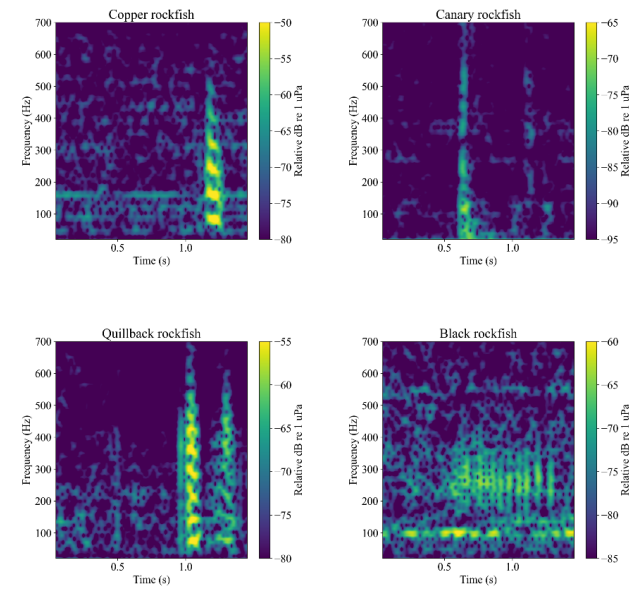
Knocks

Species	n	Knock high frequency (Hz)	Knock low frequency (Hz)	Knock peak frequency (Hz)	Knock bandwidth (Hz)	Knock duration (s)
Black rockfish	28	526.8 ± 136.3	183.4 ± 66.0	341.5 ± 129.3	345.3 ± 104.2	0.1 ± 0.0
Canary rockfish	253	756.4 ± 126.3	104.6 ± 62.1	356.2 ± 165.4	649.3 ± 133.0	0.1 ± 0.0
Copper rockfish	147	590.5 ± 133.4	40.3 ± 28.7	190.0 ± 120.2	550.3 ± 142.1	0.1 ± 0.0
Lingcod	52	694.6 ± 111.3	68.2 ± 66.9	308.2 ± 111.5	627.1 ± 147.2	0.1 ± 0.0
Pile Perch	15	653.8 ± 154.2	47.6 ± 33.9	269.3 ± 162.5	606.4 ± 146.7	0.2 ± 0.0
Quillback rockfish	44	569.9 ± 101.8	66.2 ± 51.2	233.0 ± 124.6	504.2 ± 134.1	0.1 ± 0.0
Vermillion rockfish	2	159.8 ± 22.6	37.0 ± 5.6	85.9 ± 22.0	121.3 ± 27.6	0.2 ± 0.0



Grunts

Species	n	Grunt high frequency (Hz)	Grunt low frequency (Hz)	Grunt peak frequency (Hz)	Grunt bandwidth (Hz)	Grunt duration (s)
Black rockfish	45	525.5 ± 95.4	182.6 ± 55.4	361.8 ± 102.3	342.3 ± 77.1	1.2 ± 0.9
Canary rockfish	2	643.2 ± 35.8	35.7 ± 0.0	39.1 ± 0.0	609.6 ± 33.2	0.3 ± 0.0
Copper rockfish	50	540.5 ± 131.7	33.2 ± 16.3	107.5 ± 54.6	507.3 ± 135.7	0.4 ± 0.4
Quillback rockfish	9	627.1 ± 117.3	30.1 ± 7.7	188.5 ± 110.3	597.8 ± 114.2	0.2 ± 0.1



| Species identification in the field



Darienne
Lancaster

Can we distinguish species based on acoustic features only ?

Random Forest classification

Knocks

Knock Class	Precision	Recall	F1 Score
Black rockfish	1.00	0.88	0.93
Canary rockfish	0.91	0.95	0.93
Copper rockfish	0.68	0.93	0.78
Lingcod	0.78	0.47	0.58
Pile Perch	N/A	0.00	N/A
Quillback rockfish	0.40	0.15	0.22

Grunts

Grunt Class	Precision	Recall	F1 Score
Black rockfish	0.92	0.92	0.92
Copper rockfish	0.93	0.87	0.90
Quillback rockfish	0.67	0.80	0.73

Lancaster et al., 2025

Can PAM be useful for fisheries and conservation?

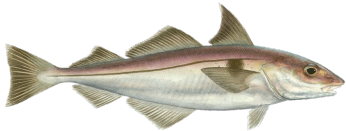
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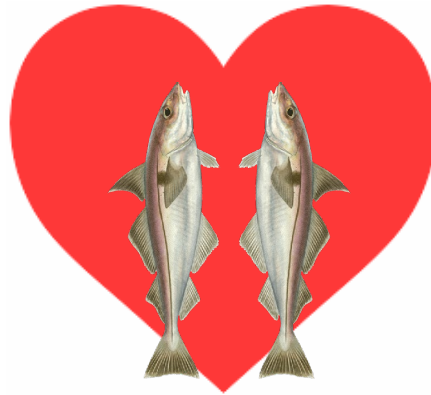
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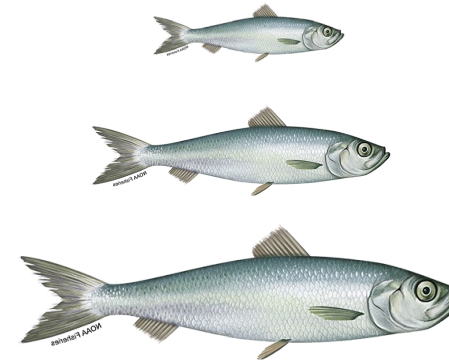
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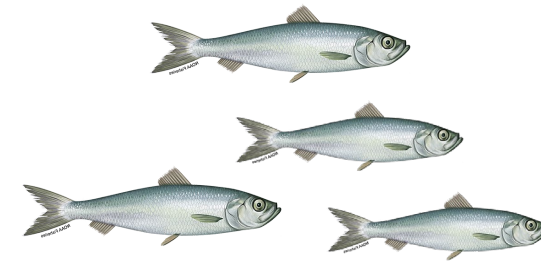
Behavior



Size



Count



Complexity

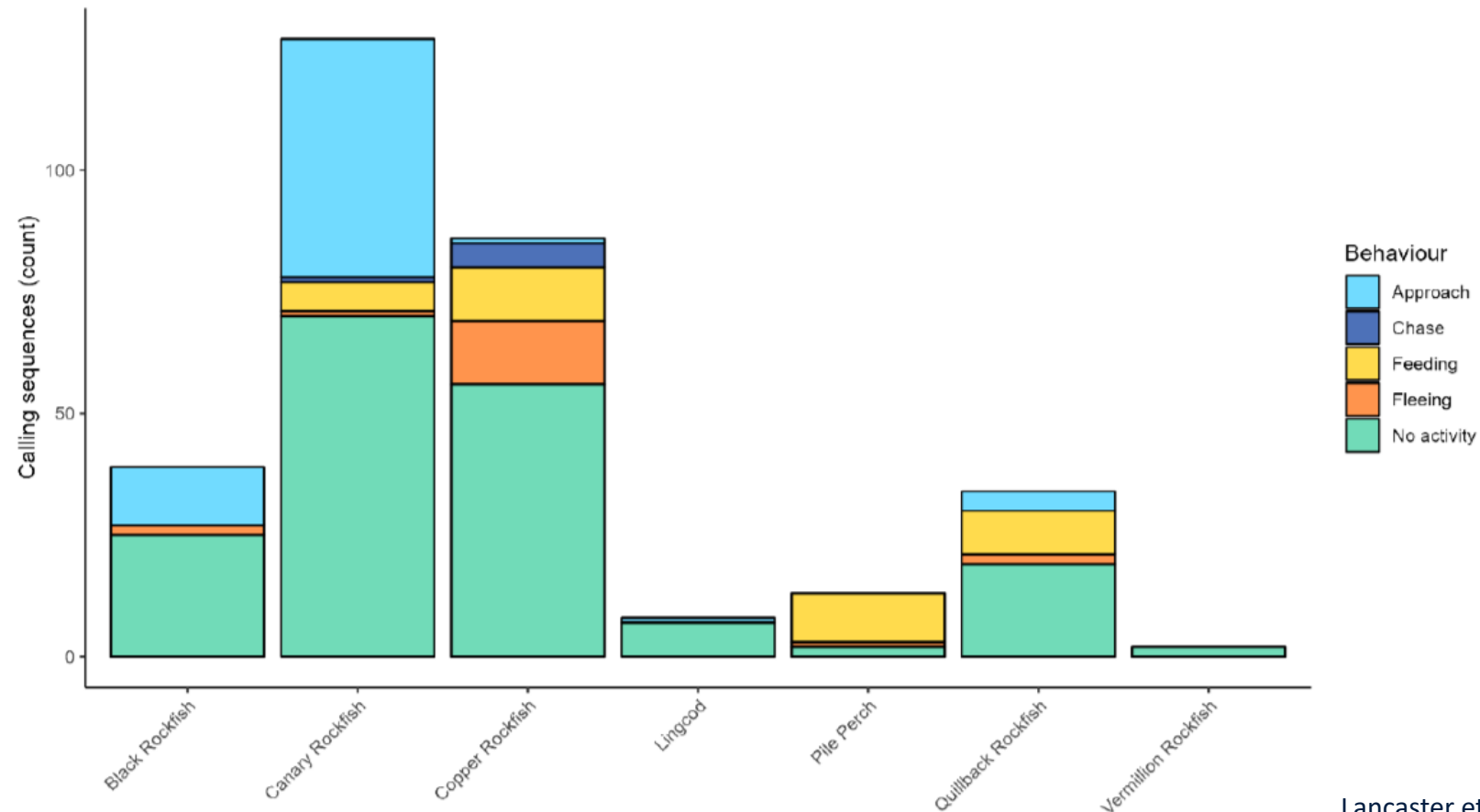


| Species-specific behavior



Darienne
Lancaster

What are the identified sounds used for ?



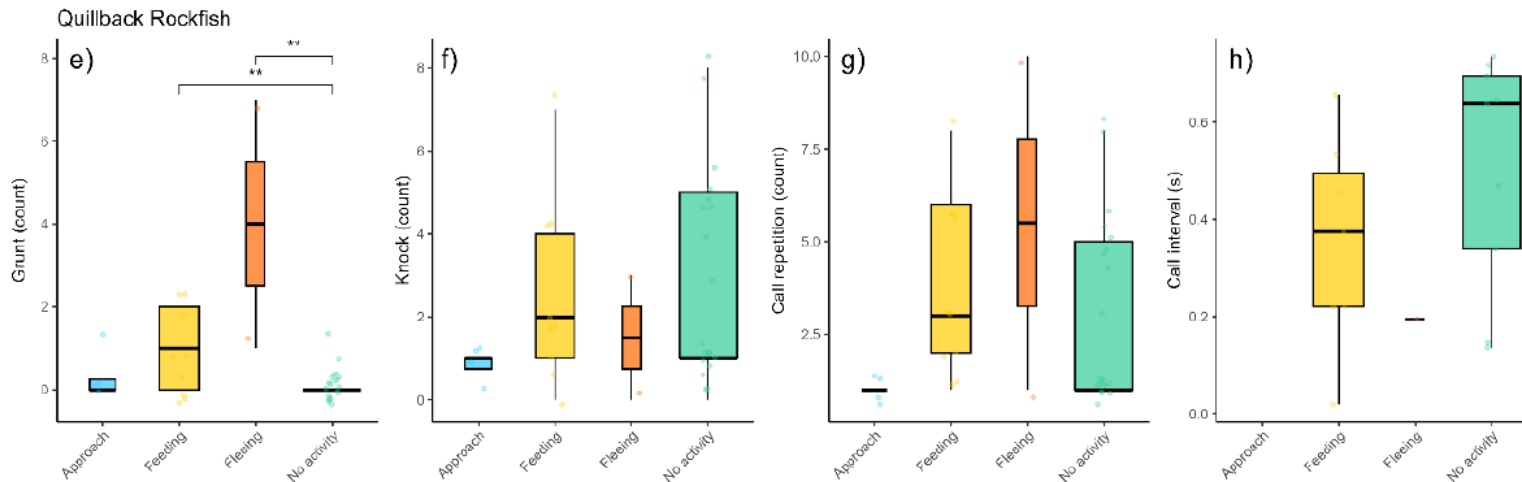
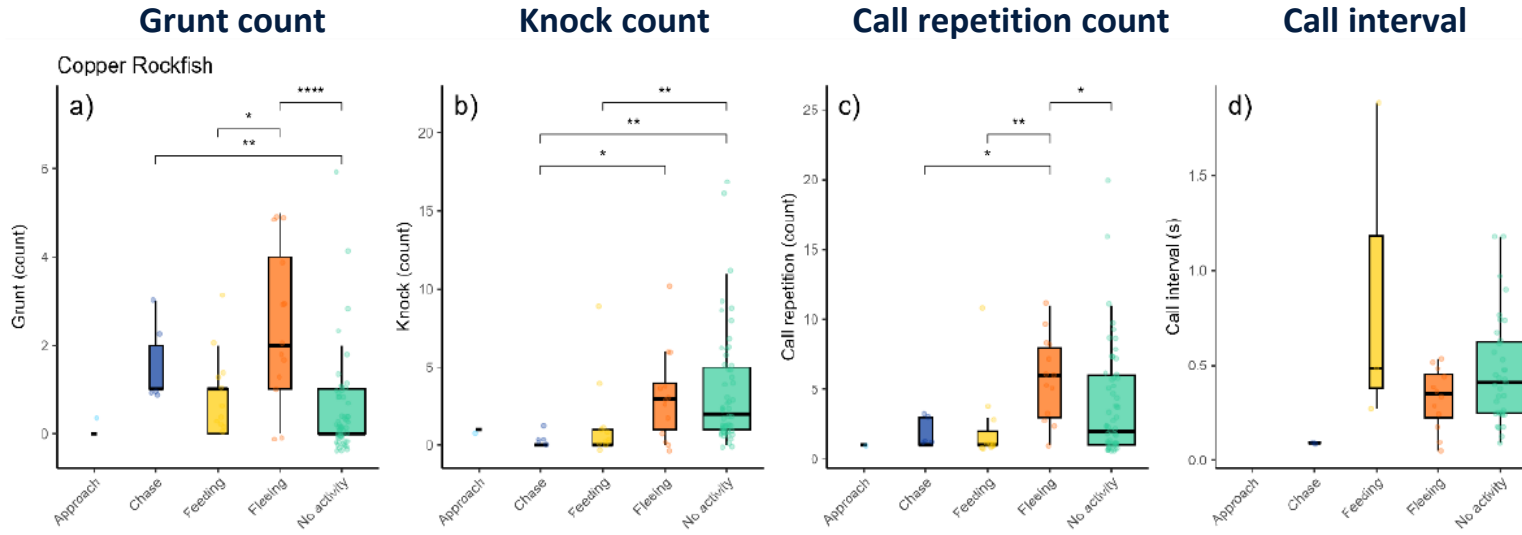
Lancaster et al., submitted

Species-specific behavior

What are the identified sounds used for ?



Darienne
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Can PAM be useful for fisheries and conservation?

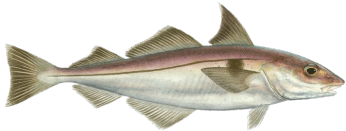
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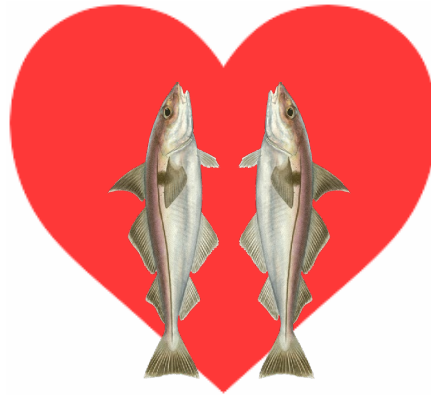
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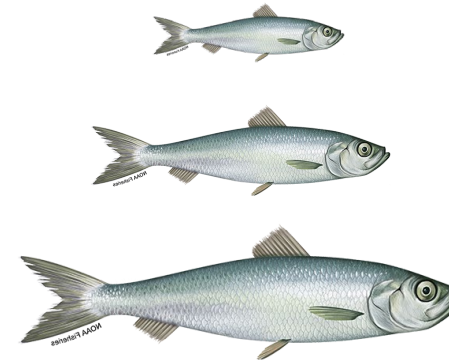
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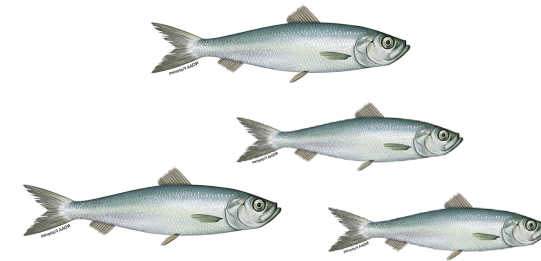
Behavior



Size



Count



Complexity

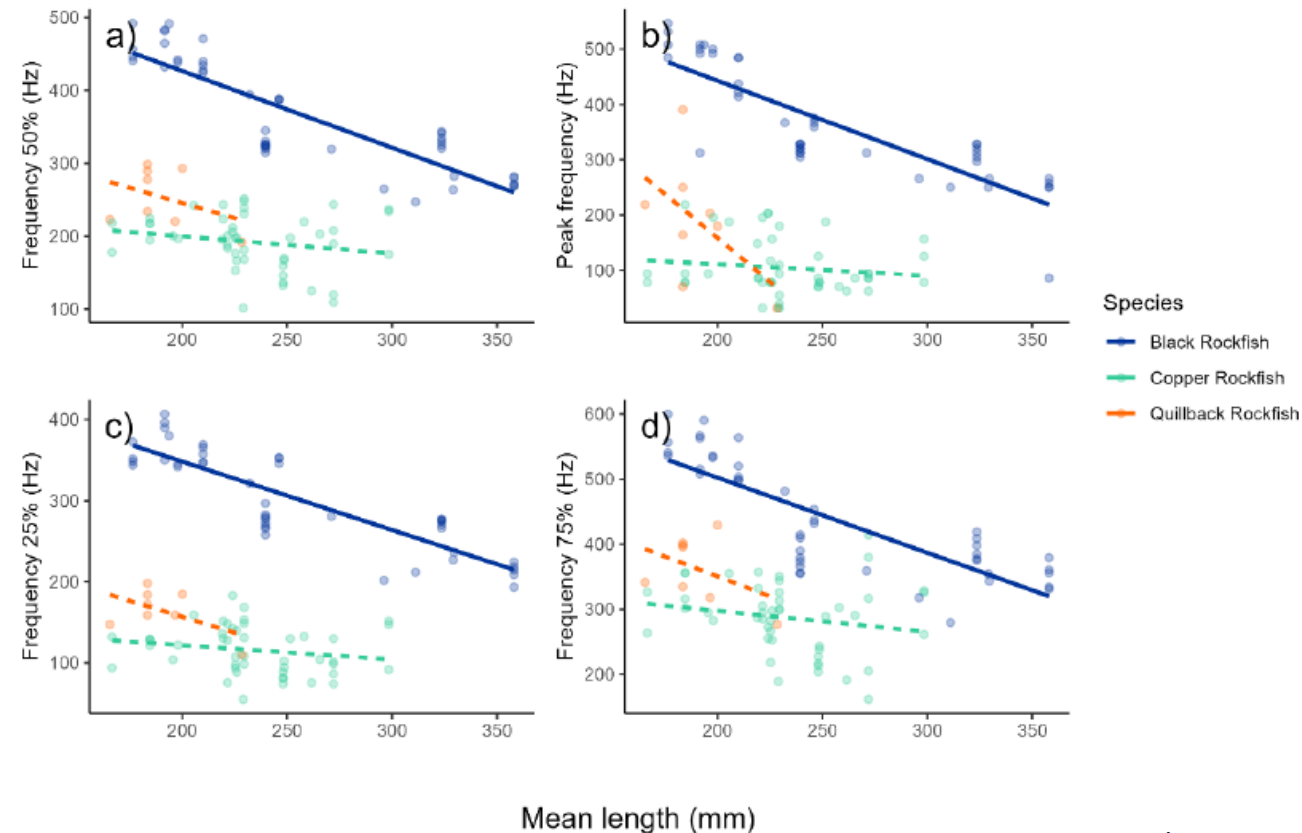
Sound features and fish size



Darienne
Lancaster

Does the acoustic signal carry information about fish size ?

Grunts



Lancaster et al., submitted

Can PAM be useful for fisheries and conservation?

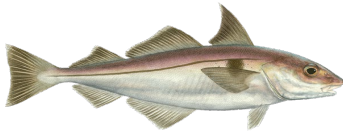
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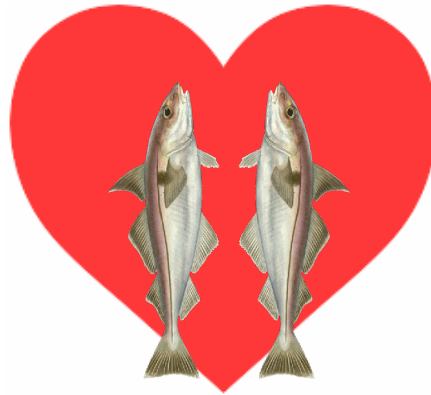
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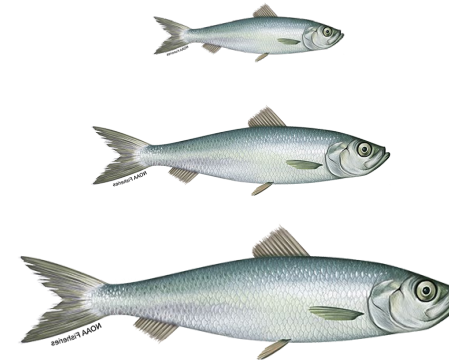
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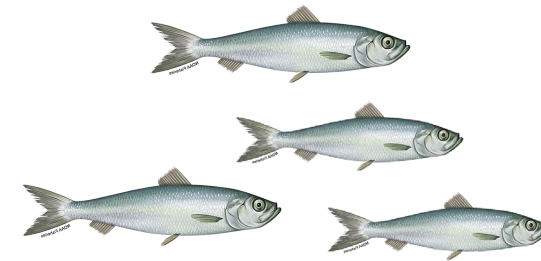
Behavior



Size



Count



Complexity



| Fish abundance

Where are we with estimating fish abundance from passive acoustics ?

- Several studies have shown that it is possible for some species (e.g. Rowell et al 2017)
- Data from A-V arrays could be used to start looking at cue rate per species
- Requires (manual?) tracking of fish in video data
- Relative abundance might be enough (Caiger et al.2020)

Can PAM be useful for fisheries and conservation?

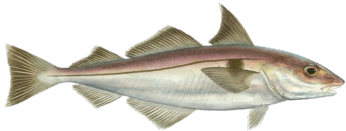
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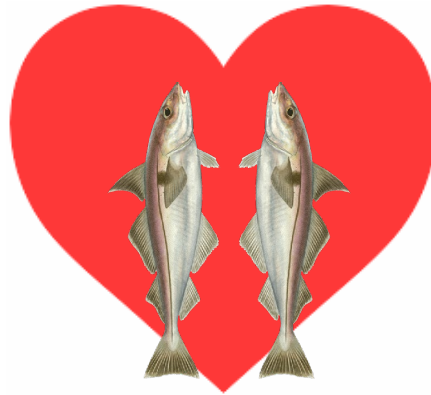
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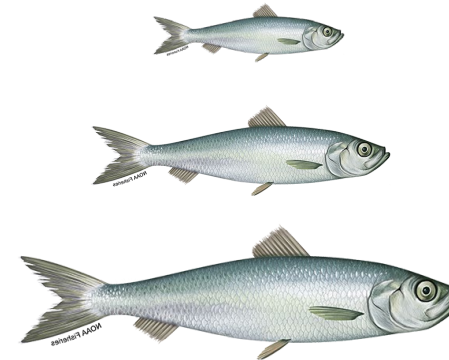
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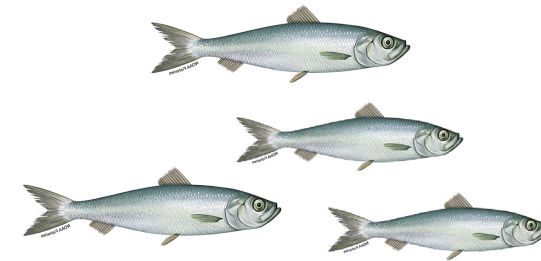
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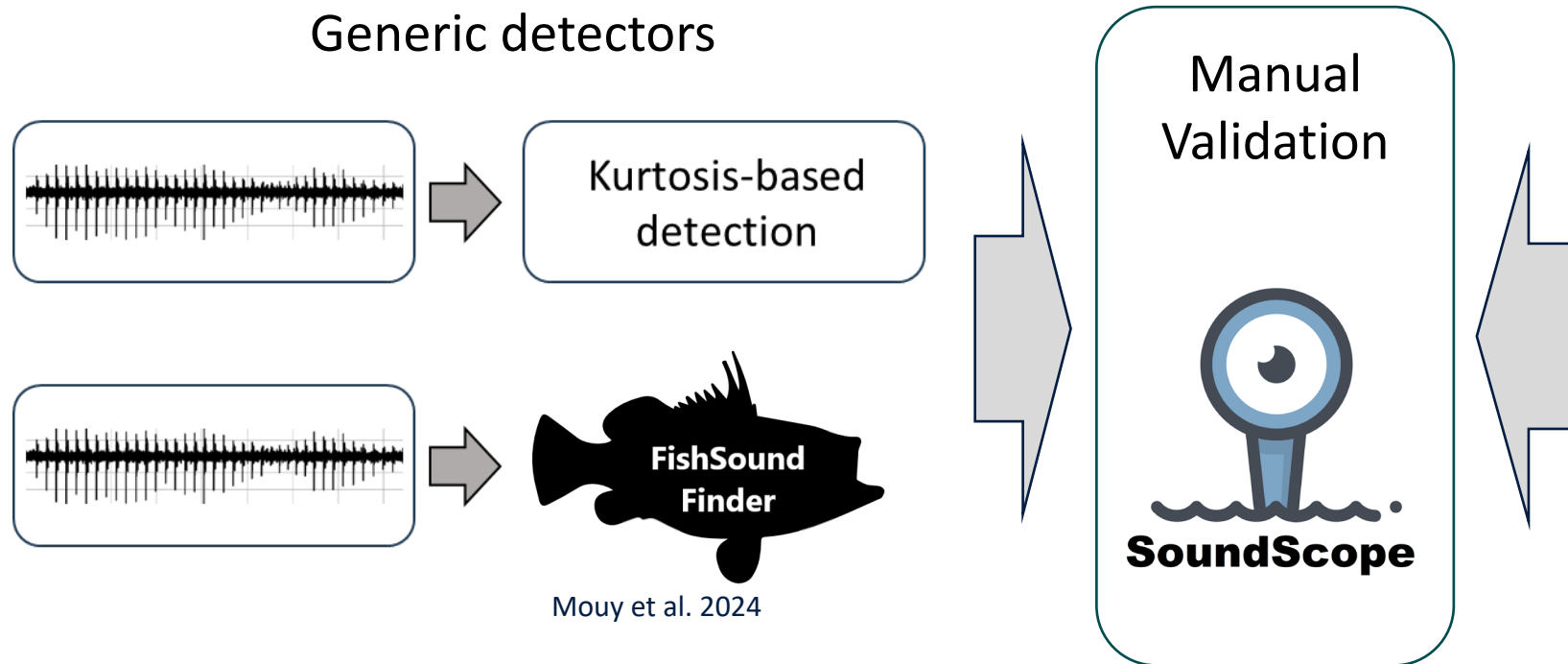
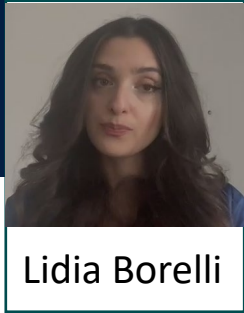


Complexity

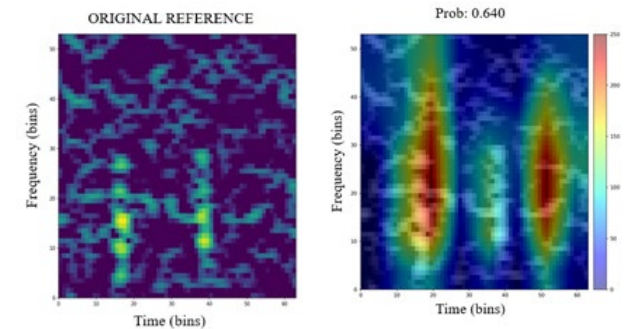
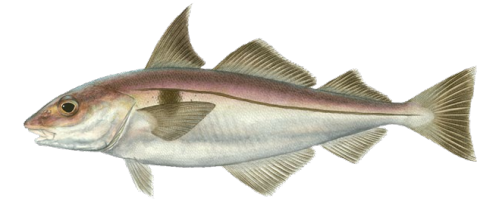


Automated processing

Generic and species-specific detectors



Haddock detector



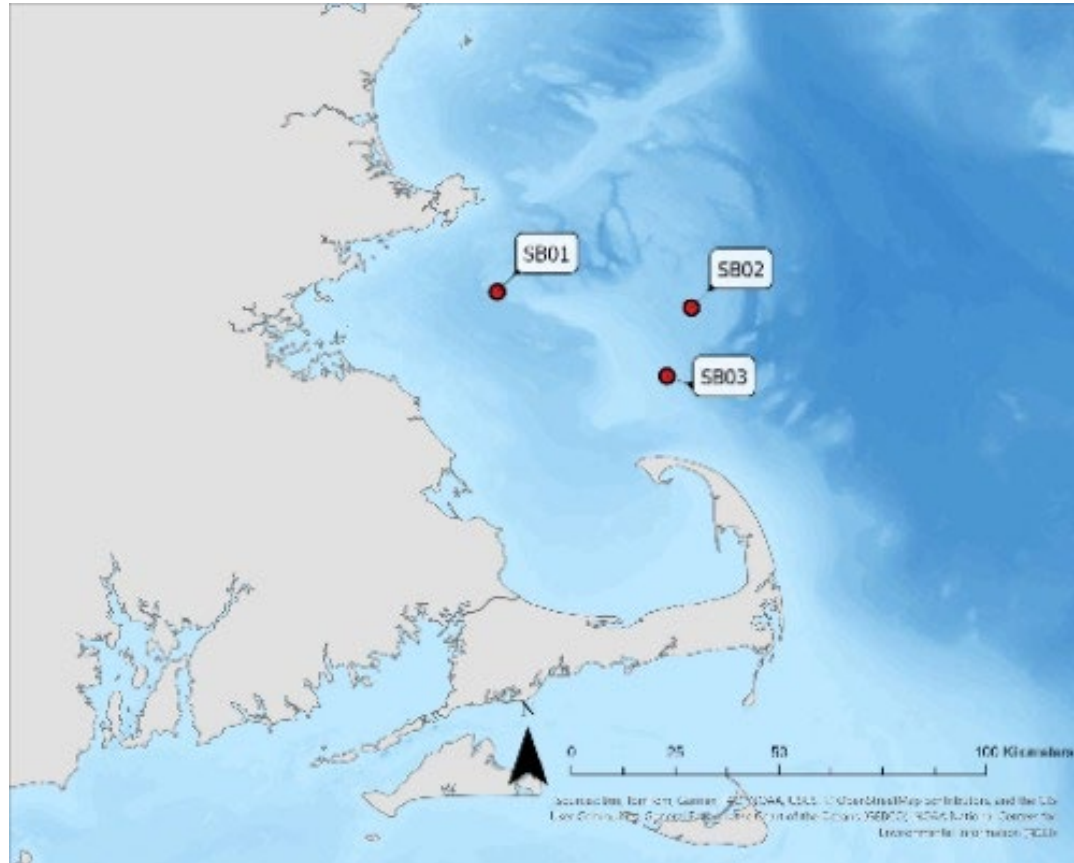
Borelli et al., in prep

Monitoring haddock with PAM

Stellwagen Bank National Marine Sanctuary



Anne
Smith



3 locations:

- SB01: 48/50 m
- SB02: 70/72 m
- SB03: 43/48 m

Analysis period

- Start: Nov. 2018
- End: Apr. 2022

Analysis approach

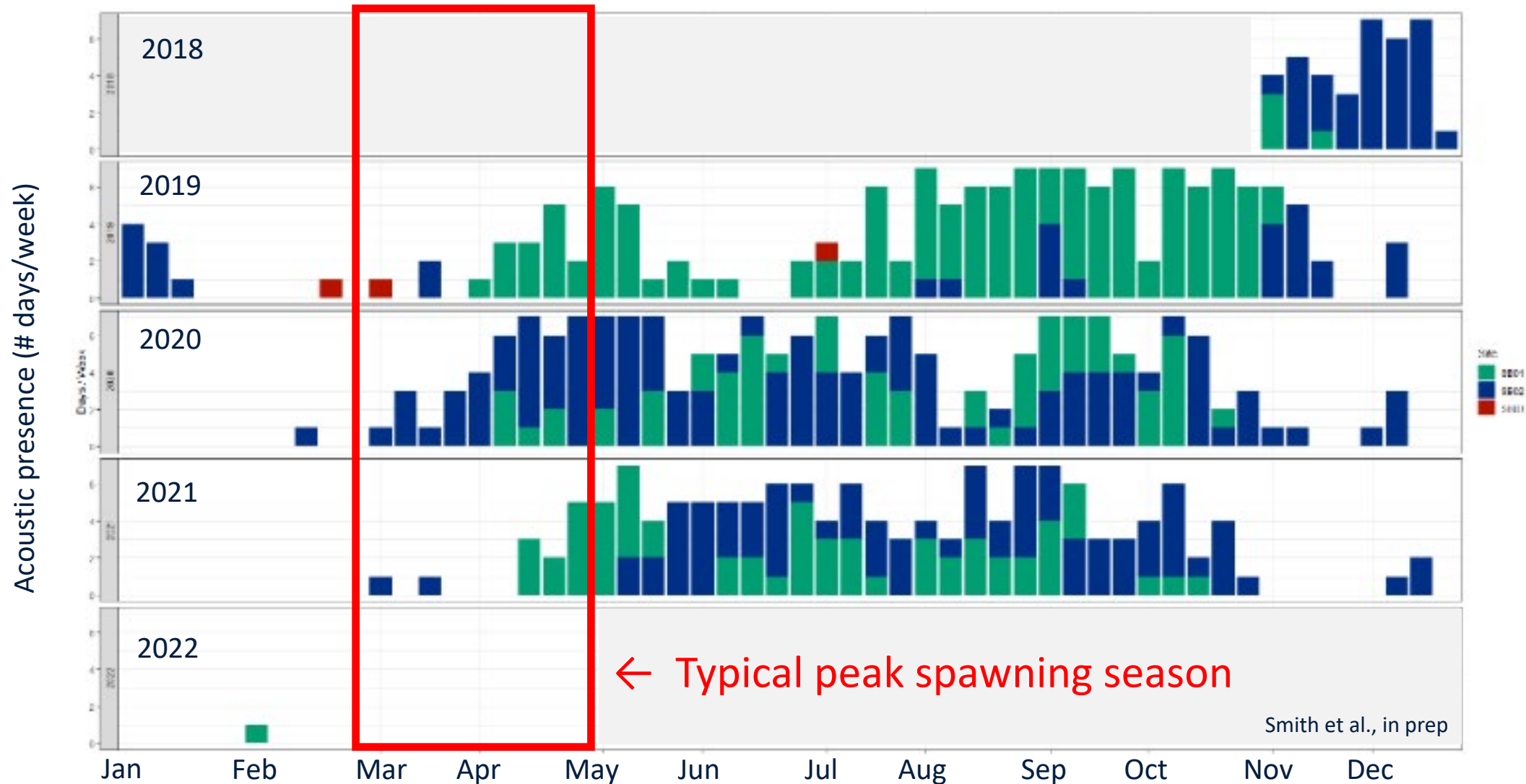
- Generic kurtosis detector
- Manual verification with SoundScope

Monitoring haddock with PAM



Anne Smith

Stellwagen Bank National Marine Sanctuary

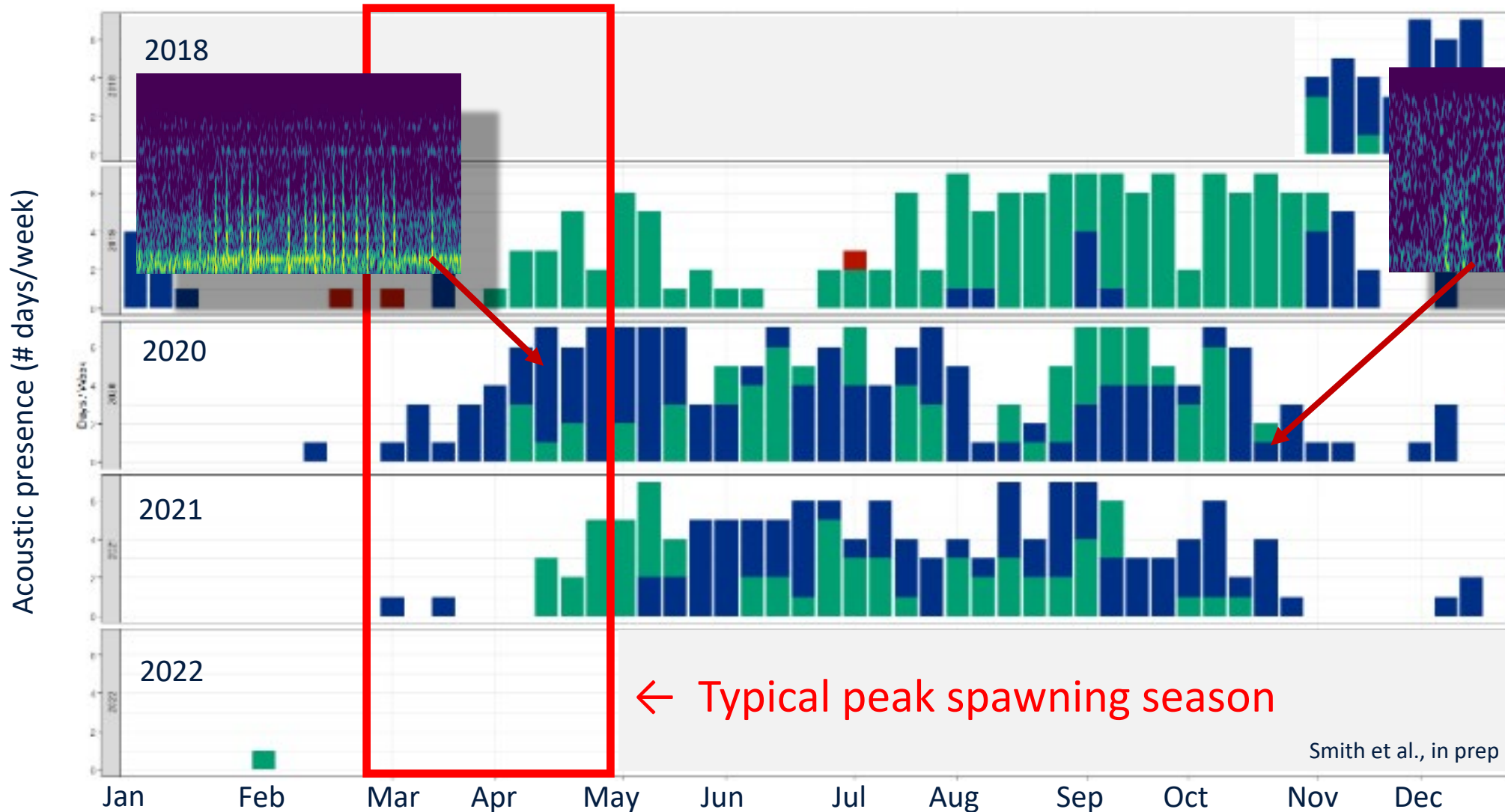


Monitoring haddock with PAM



Anne
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Stellwagen Bank National Marine Sanctuary

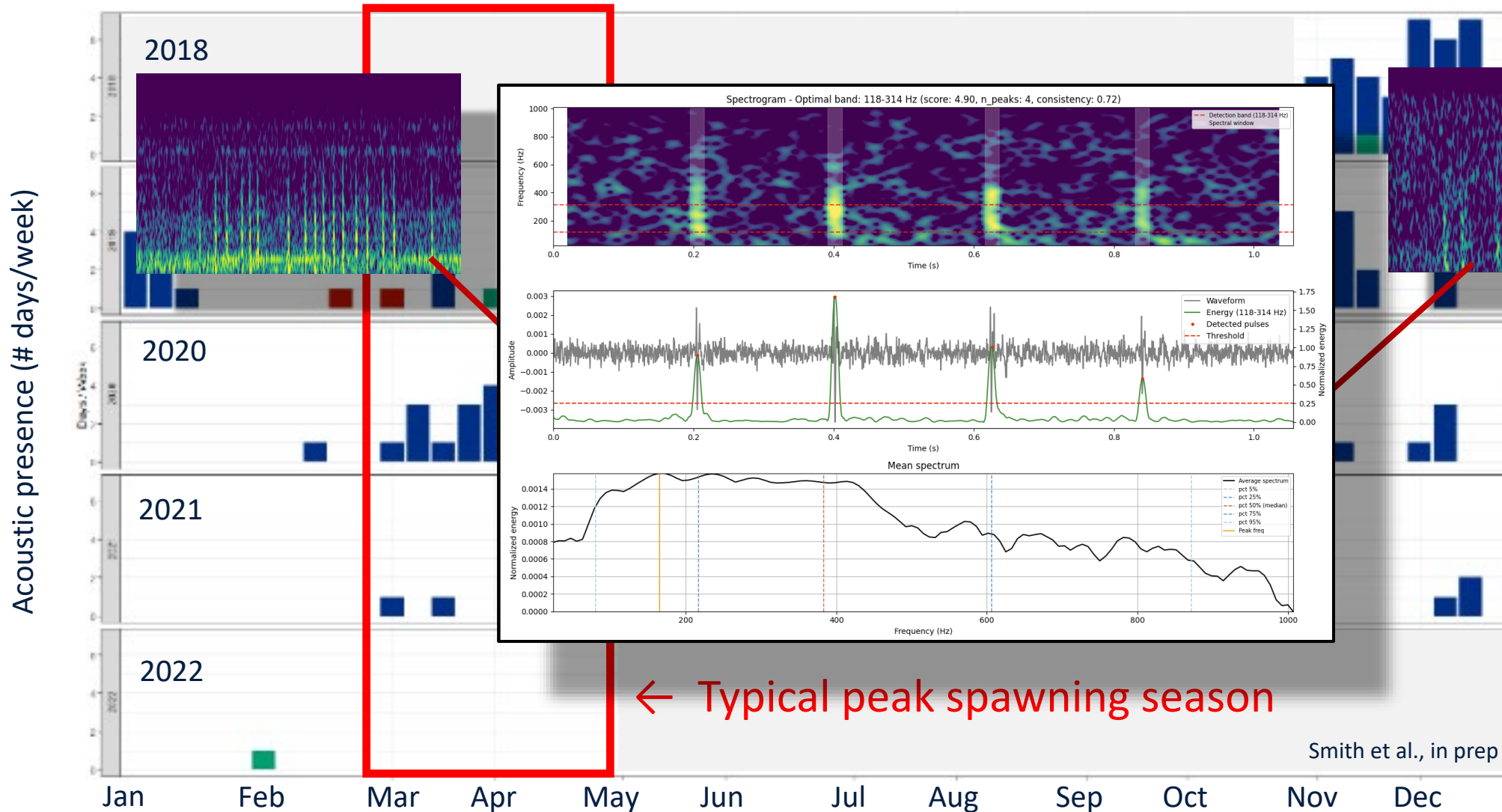


Monitoring haddock with PAM

Stellwagen Bank National Marine Sanctuary



Anne Smith



Incorporating PAM to stock assessments

Atlantic cod

- Over 10 years of PAM data from Mass Bay and 6 years at Cox Ledge in SNE
- Fixed-station and glider sampling paired with acoustic telemetry

Comparing Atlantic Cod Temporal Spawning Dynamics across a Biogeographic Boundary: Insights from Passive Acoustic Monitoring

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Institute of Marine Sciences, University of North Carolina at Chapel Hill, 3431 Arendell Street, Morehead City, North Carolina 28557, USA

Timothy J. Rowell
National Oceanic and Atmospheric Administration, 166 Water Street, Woods Hole, Massachusetts 02543

Micah J. Dean^a
Massachusetts Division of Marine Fisheries, 100 Water Street, Woods Hole, Massachusetts 02543

Aaron N. Rice^a
E. Liu Yang Center for U.S. Marine Research, 100 Water Street, Woods Hole, Massachusetts 02543

Sofie M. Van Parijs^a
National Oceanic and Atmospheric Administration, 166 Water Street, Woods Hole, Massachusetts 02543

ICES Journal of Marine Science

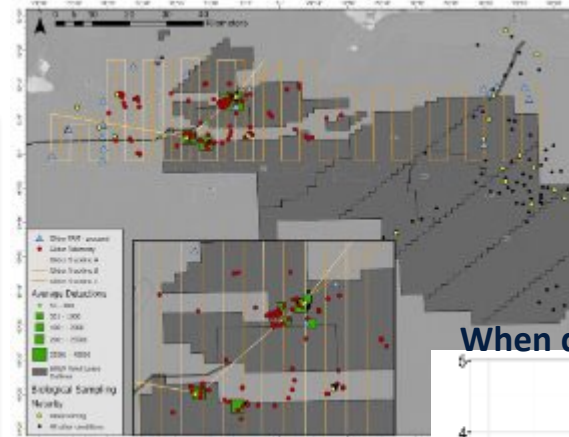
ICES Journal of Marine Science (2019), 78(1), 1600–1625. doi:10.1093/icesjms/ksz064

Original Article

Identifying the distribution of Atlantic cod spawning using multiple fixed and glider-mounted acoustic technologies

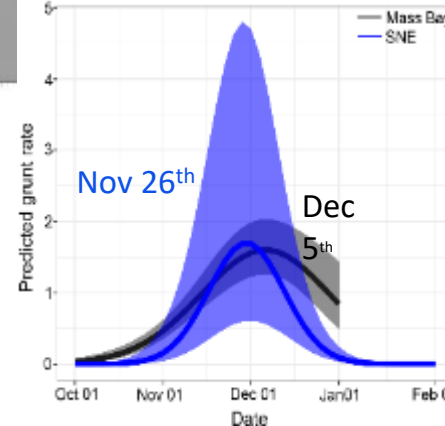
Douglas R. Zemeckis^{1,2,3}, Micah J. Dean², Annamaria I. DeAngelis^{1,4}, Sofie M. Van Parijs¹, William S. Hoffman², Mark F. Baumgartner², Leila T. Hatch², Steven X. Cadrin¹, and

Where did we hear them?

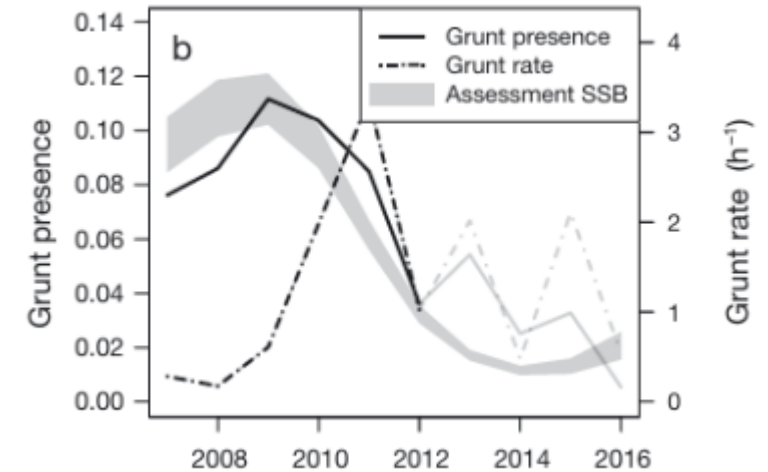


Van Hoeck et al. (2023)

When did we hear them?



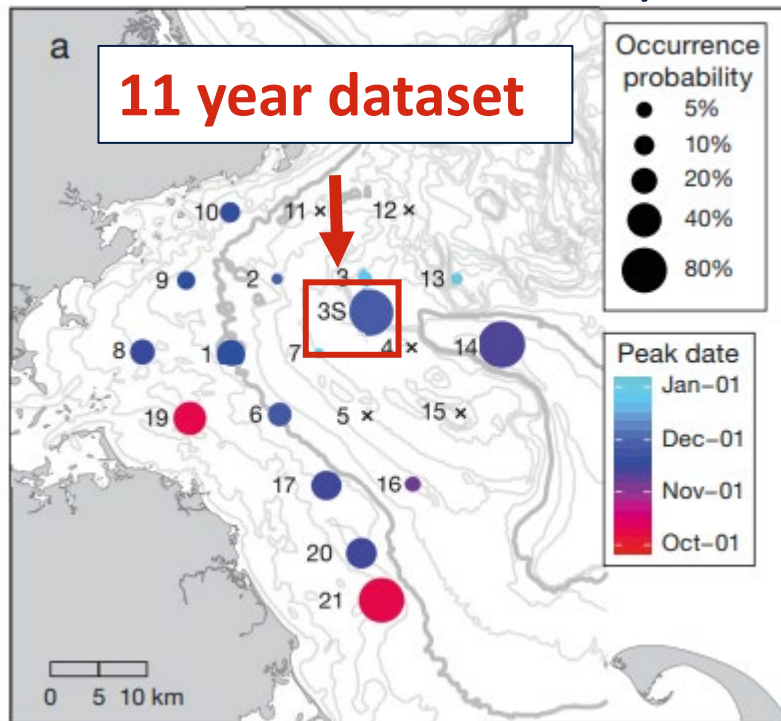
Fishery Management Connections: PAM & Spawning Stock Biomass Trends



Caiger et al. (2020)

Incorporating PAM to stock assessments

Stellwagen Bank National Marine Sanctuary



Caiger et al. (2020)

- NEFSC cross-division collaboration to integrate Passive Acoustic data into the Western Gulf of Maine Atlantic cod and Gulf of Maine Haddock Stock Assessments
 - Becca Van Hoeck, Annie Smith, Rhett Finley, Amanda Holdman, Passive Acoustic Branch
 - Conor McManus, Marine Development and Advanced Technology Branch,
 - Charles Peretti, Resource Evaluation and Assessment Population Dynamics
 - Xavier Mouy, WHOI
- Next steps:
 - Complete the 11 year timeseries of Atlantic cod occurrence in SBNMS
 - Model the probability of grunt occurrence and rate
 - Compare the PAM trend to spawning stock biomass
 - Socialize the data with the Council
 - Identify quantitative measures that could be incorporated in to stock assessment model

| Conclusions

Advances made:

- Instrumentation
- Processing techniques
- Manual analysis interfaces
- Tool/data accessibility (open source)
- Better understanding of (some) fish sounds

Challenges remaining:

- Need more accessible/cheaper) instrumentation
- Still data gaps in low visibility conditions / at night
- Automated detectors still take long time to train/evaluate
- Defining best sampling design for stock assessments applications

Relevance for other regions:

- Deployment of AV-instrumentation (if visibility allows it)
- Rockfish dataset (West coast only)
- Some of the analysis software (generic detectors, SoundScope)